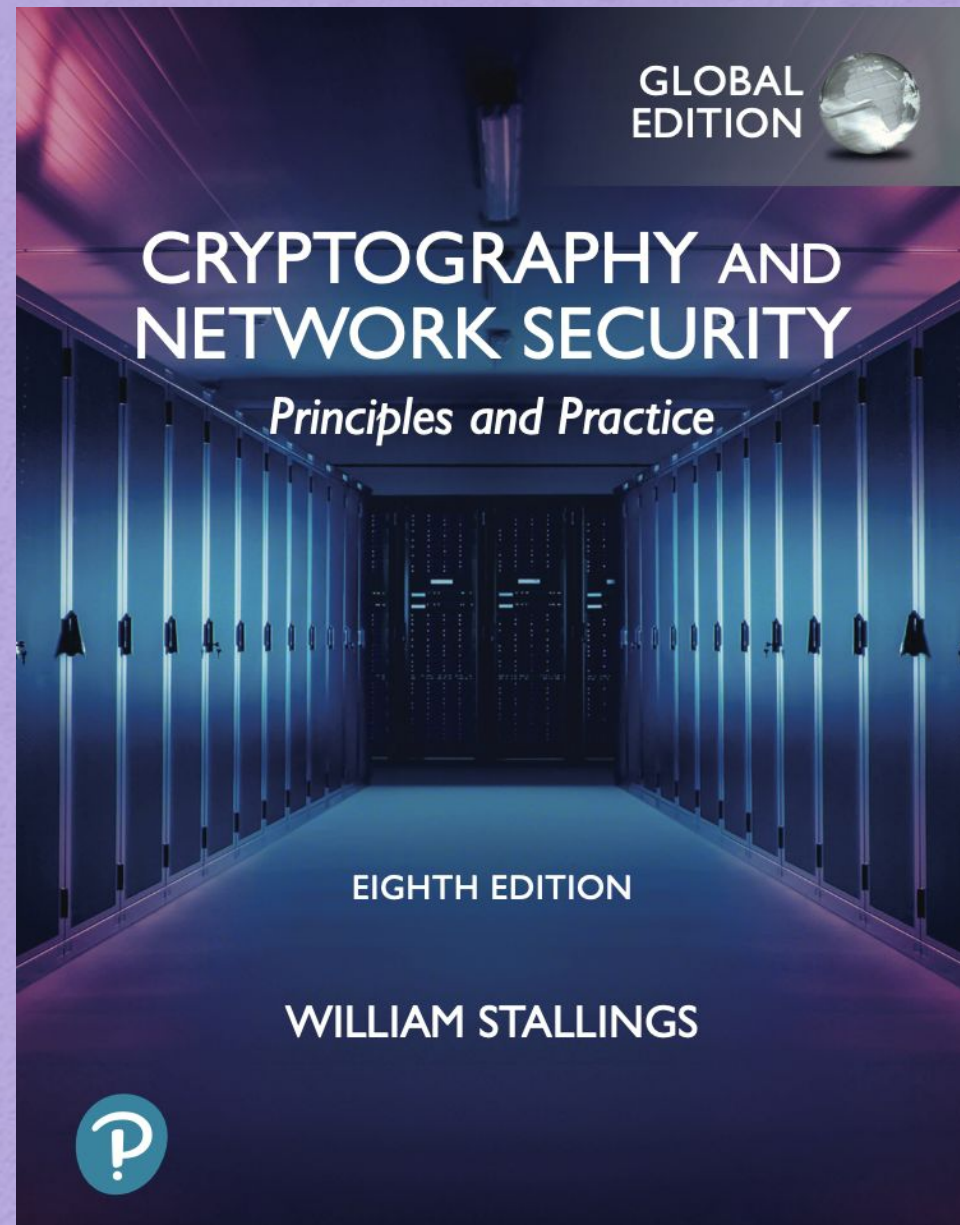


University of Nevada – Reno  
Computer Science & Engineering  
Department

CS454/654 Reliability and Security  
of Computing Systems - Fall 2024



## BLOCK CIPHER OPERATION

### 7.1 Multiple Encryption and Triple DES

- Double DES
- Triple DES with Two Keys
- Triple DES with Three Keys

### 7.2 Electronic Codebook

### 7.3 Cipher Block Chaining Mode

### 7.4 Cipher Feedback Mode

### 7.5 Output Feedback Mode

### 7.6 Counter Mode

### 7.7 XTS-AES Mode for Block-Oriented Storage Devices

- Tweakable Block Ciphers
- Storage Encryption Requirements
- Operation on a Single Block
- Operation on a Sector

### 7.8 Format-Preserving Encryption

- Motivation
- Difficulties in Designing an FPE
- Feistel Structure for Format-Preserving Encryption
- NIST Methods for Format-Preserving Encryption

### 7.9 Key Terms, Review Questions, and Problems



## 7.1 MULTIPLE ENCRYPTION AND TRIPLE DES

DES (Data Encryption Standards) is vulnerable to brute force attacks.

**Table 4.5** Average Time Required for Exhaustive Key Search

| Key Size (bits)             | Cipher         | Number of Alternative Keys           | Time Required at $10^9$ Decryptions/s           | Time Required at $10^{13}$ Decryptions/s |
|-----------------------------|----------------|--------------------------------------|-------------------------------------------------|------------------------------------------|
| 56                          | DES            | $2^{56} \approx 7.2 \times 10^{16}$  | $2^{55}$ ns = 1.125 years                       | 1 hour                                   |
| 128                         | AES            | $2^{128} \approx 3.4 \times 10^{38}$ | $2^{127}$ ns = $5.3 \times 10^{21}$ years       | $5.3 \times 10^{17}$ years               |
| 168                         | Triple DES     | $2^{168} \approx 3.7 \times 10^{50}$ | $2^{167}$ ns = $5.8 \times 10^{33}$ years       | $5.8 \times 10^{29}$ years               |
| 192                         | AES            | $2^{192} \approx 6.3 \times 10^{57}$ | $2^{191}$ ns = $9.8 \times 10^{40}$ years       | $9.8 \times 10^{36}$ years               |
| 256                         | AES            | $2^{256} \approx 1.2 \times 10^{77}$ | $2^{255}$ ns = $1.8 \times 10^{60}$ years       | $1.8 \times 10^{56}$ years               |
| 26 characters (permutation) | Monoalphabetic | $2! = 4 \times 10^{26}$              | $2 \times 10^{26}$ ns = $6.3 \times 10^9$ years | $6.3 \times 10^6$ years                  |

Completely new algorithm that is resistant to both cryptanalytic and brute-force attacks, of which AES is a prime example. *NIST deprecated both DES/3DES*

Continue investment in existing approach, such as using multiple encryption with DES and multiple keys.

## 7.1 MULTIPLE ENCRYPTION AND TRIPLE DES

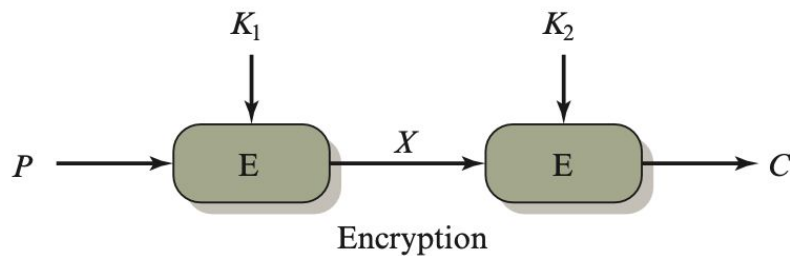
### Double DES

Given a plaintext **P** and two encryption keys **K<sub>1</sub>** and **K<sub>2</sub>**, ciphertext **C** is generated as

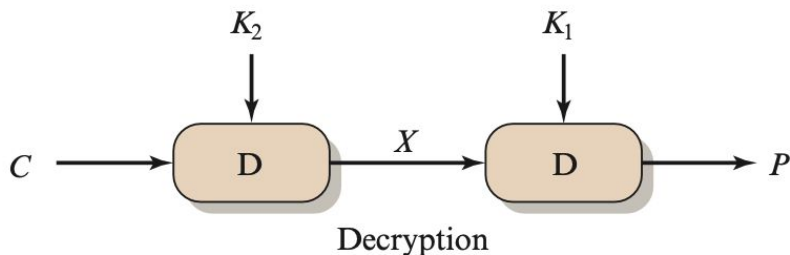
$$C = E(K_2, E(K_1, P))$$

Decryption requires that the keys be applied in **reverse** order:

$$P = D(K_1, D(K_2, C))$$



*2nd stage*



**(a) Double encryption**



## 7.1 MULTIPLE ENCRYPTION AND TRIPLE DES

Suppose it were true for DES, for all 56-bit key values, that given any two  
With  $2^{64}$  possible inputs, the number of different mapping is

$$(2^{64})! = 10^{34738000000000000000} > (10^{10^{20}})$$

DES defines **one mapping** for **each different key**, and a total number of mappings is

$$2^{56} < 10^{17}$$

↓                      ↓

AES-256      AES-128

Therefore, it is reasonable to assume that if DES **is used twice** with different keys, **it will produce one of the many mappings** that are **not defined by a single application of DES**.

## 7.1 MULTIPLE ENCRYPTION AND TRIPLE DES

### Meet in the Middle Attack

$$C = E(K_2, E(K_1, P))$$

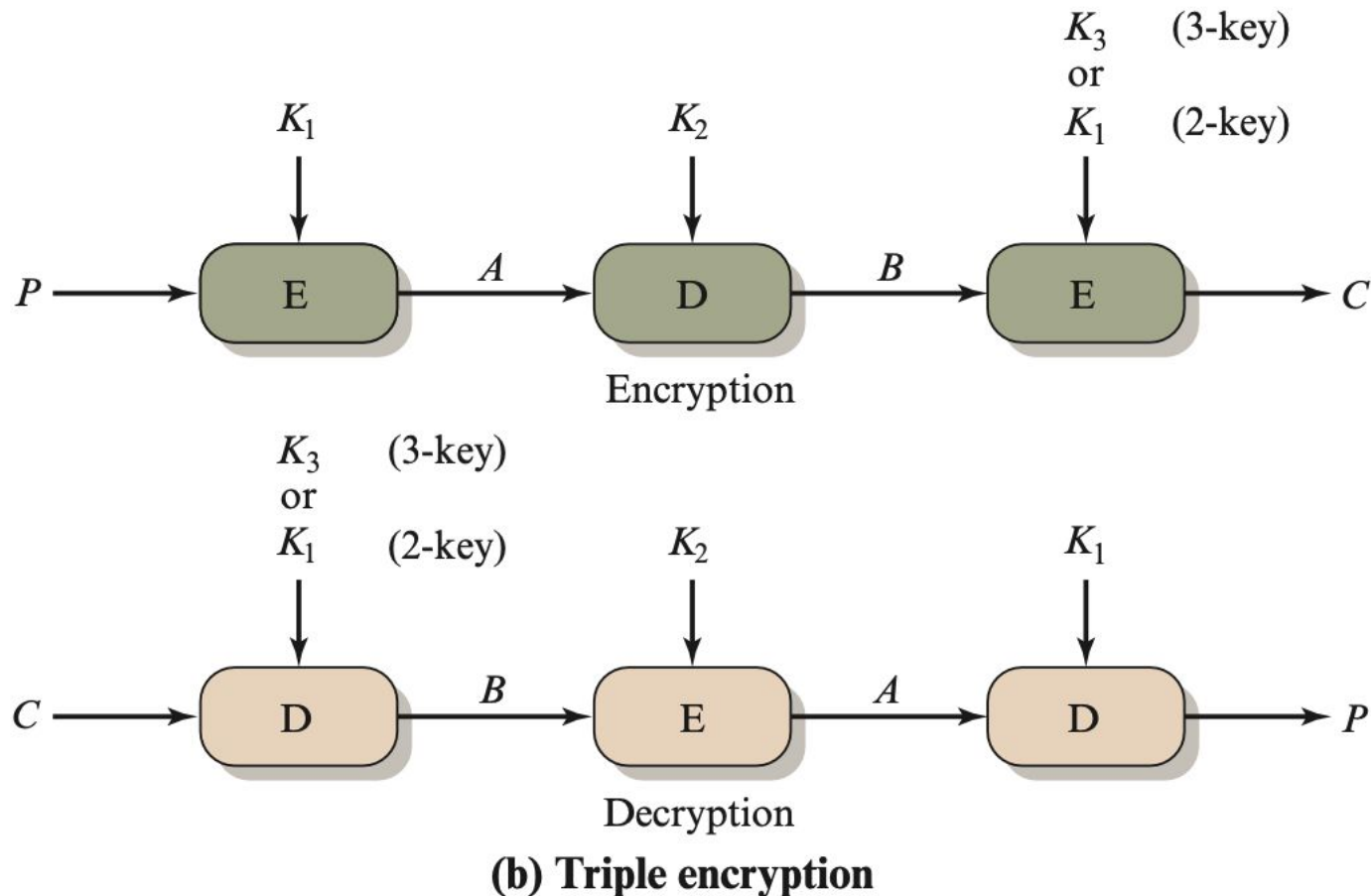
$$X = E(K_1, P) = D(K_2, C)$$

First, **encrypt**  $P$  for all  $2^{56}$  possible values of  $K_1$ . Store these results in a table and then sort the table by the values of  $X$ . Next, **decrypt**  $C$  using all  $2^{56}$  possible values of  $K_2$ . As each decryption is produced, check the result against the table for a match.

The result is that a **known plaintext attack** will **succeed** against double DES, with an effort on the **order of**  $2^{56}$ , which is not much more than the  $2^{55}$  required for single DES.



## 7.1 MULTIPLE ENCRYPTION AND TRIPLE DES



**Figure 7.1** Multiple Encryption

<sup>2DES</sup>  
two versions of 3DES; one using two keys and one using three keys.

## 7.1 MULTIPLE ENCRYPTION AND TRIPLE DES

*Sweet32?*

- Currently, there are **no practical cryptanalytic attacks** on 3DES.
- Couple attacks techniques mentioned in book but they are not practical.
  - In general proposed approaches try to reduce the 3DES to 2DES.  
*why? simplicity?*

**Table 4.5** Average Time Required for Exhaustive Key Search

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# Modes of Operation

- To apply a block cipher in a variety of applications, there are five modes of operation.
  - In essence, a mode of operation is a technique for enhancing the effect of a cryptographic algorithm or adapting the algorithm for an application, such as applying a block cipher to a sequence of data blocks or a data stream.
  - These modes are intended for use with any symmetric block cipher, including triple DES and AES.
1. Electronic Codebook (ECB)
  2. Cipher Block Chaining (CBC)
  3. Cipher Feedback (CFB)
  4. Output Feedback (OFB)
  5. Counter (CTR)

# ELECTRONIC CODEBOOK

- The **simplest mode**, in which plaintext is **handled one block at a time** and each block of plaintext is encrypted using the same key. *of size  $n$  bits*
- The term **codebook** is used because, **for a given key**, there is a **unique ciphertext for every  $b$ -bit block of plaintext**. Thus, resembles a gigantic codebook in which there is an entry for every possible  $b$ -bit plaintext pattern showing its corresponding ciphertext.
- For a message **longer than  $b$  bits**, split into  $b$ -bit blocks and pad the last if necessary. *add zeroes?*
- The ECB mode **should be used only to secure messages shorter than a single block** such as to encrypt a secret key.
- For lengthy messages, it **may not be secure**. If the message is highly structured, it may be possible for a cryptanalyst to exploit these regularities. *frequency analysis!*



# CIPHER BLOCK CHAINING MODE

- Technique in which the same plaintext block, if repeated, produces different ciphertext blocks.

- In this scheme, the **input** to the encryption algorithm is the **XOR** of the **current plaintext block and the preceding ciphertext block**; the same key is used for each block.

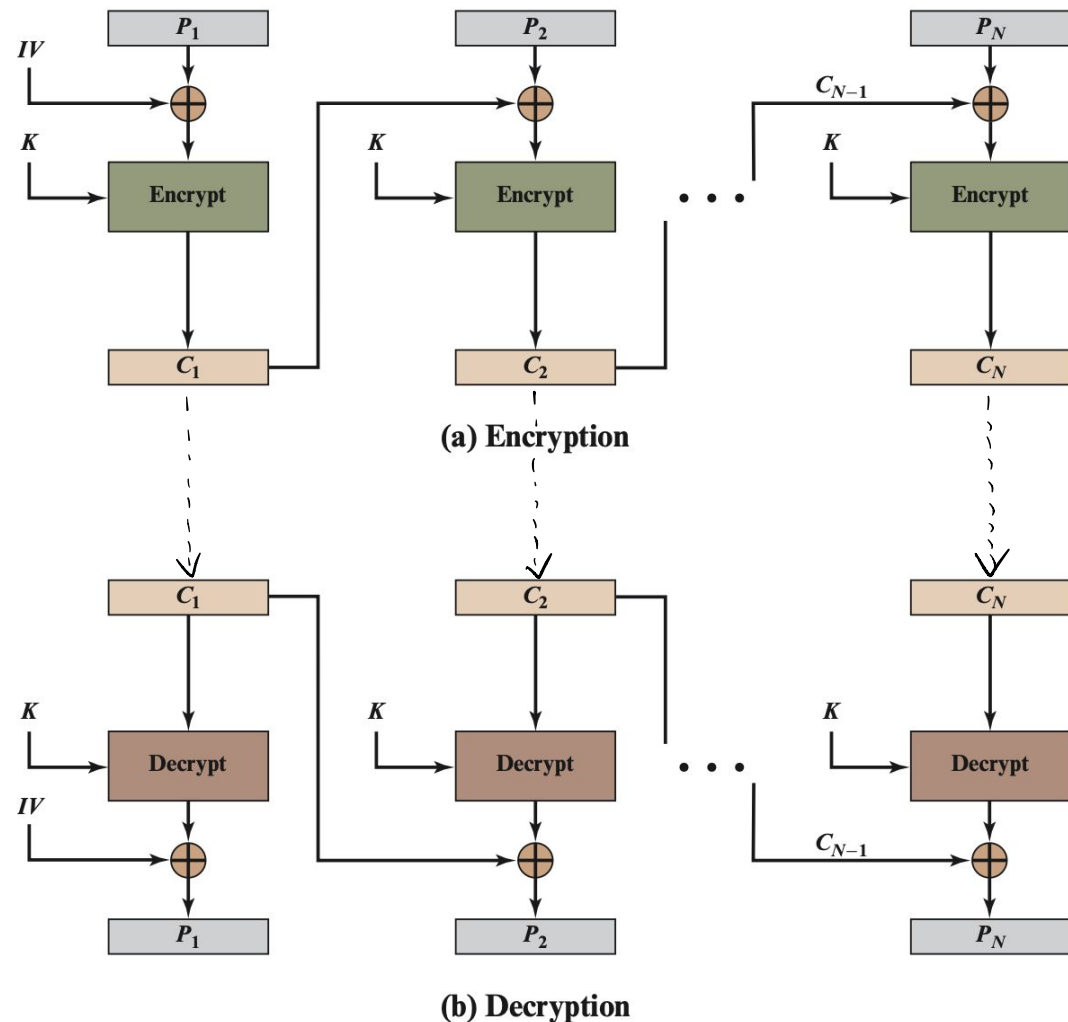
$\oplus$

→ prev. data affects current data encryption

- CBC mode requires that the last block be padded to a full  $b$  bits if it is a partial block.  
w/ zeroes?

# CIPHER BLOCK CHAINING MODE

IV is **nonce**:  
counter, a  
timestamp, or a  
message  
number.



**Figure 7.4** Cipher Block Chaining (CBC) Mode



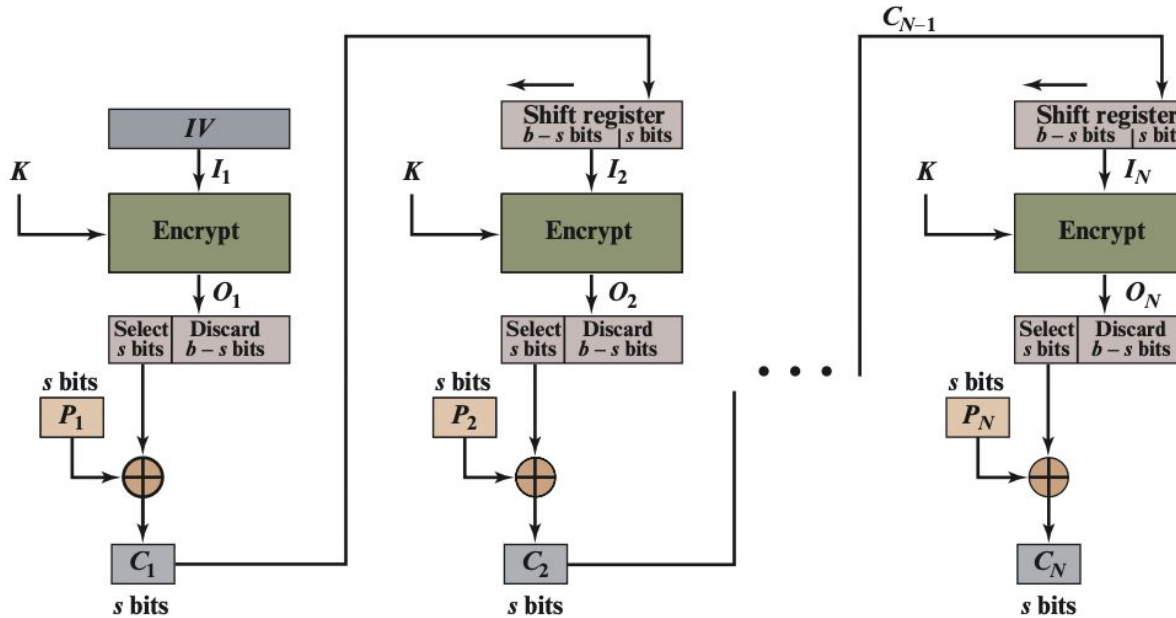
# CIPHER

## FEEDBACK MODE

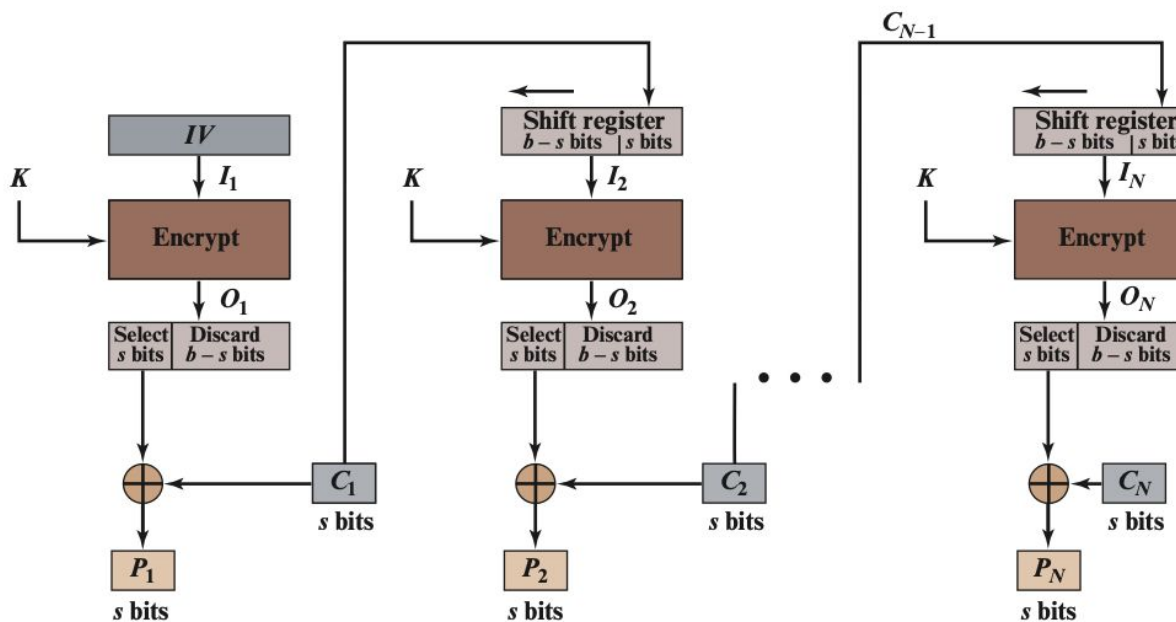
Unit of transmission is **s** bits;  
a common value is  $s = 8$ .

Rather than **blocks of b bits**,  
the plaintext is divided into  
**segments of s bits**.

what?



(a) Encryption

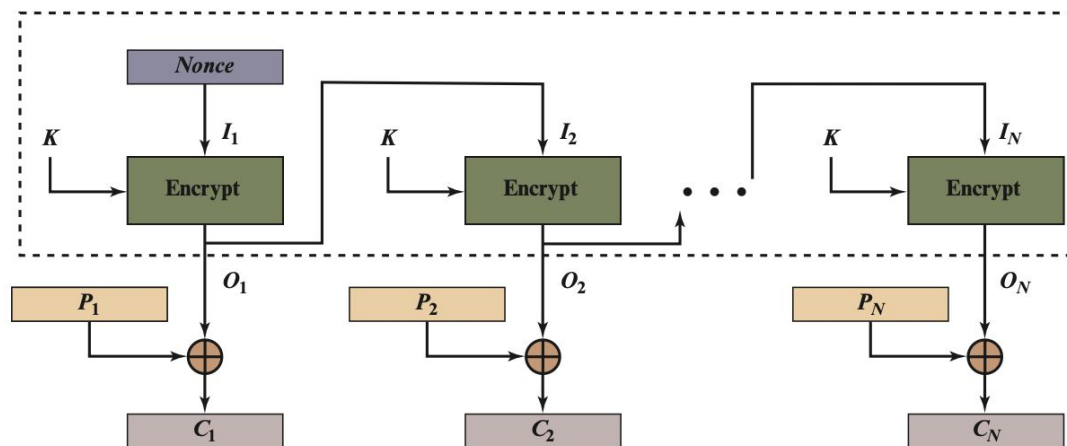


(b) Decryption

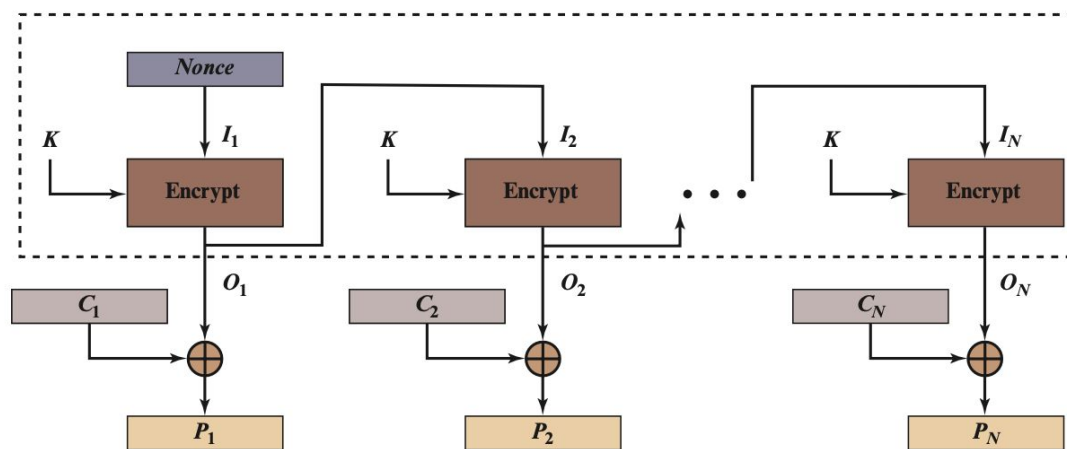
Figure 7.5 s-bit Cipher Feedback (CFB) Mode

# OUTPUT FEEDBACK MODE (OFB)

One **advantage** of the OFB method is that **bit errors** in transmission **do not propagate**. For example, if a bit error occurs in  $C_1$ , only the recovered value of  $P_1$  is affected; subsequent plaintext units are not corrupted.



(a) Encryption



(b) Decryption

Figure 7.6 Output Feedback (OFB) Mode



# COUNTER MODE (CTR)

Although interest in the counter (CTR) mode has increased recently with applications to ATM (asynchronous transfer mode) network security and IPsec, this mode was proposed in 1979.

A counter equal to the plaintext block size is used, it must be different for each plaintext block that is encrypted.

Typically, the counter is initialized to some value and then **incremented by 1** for each subsequent block.

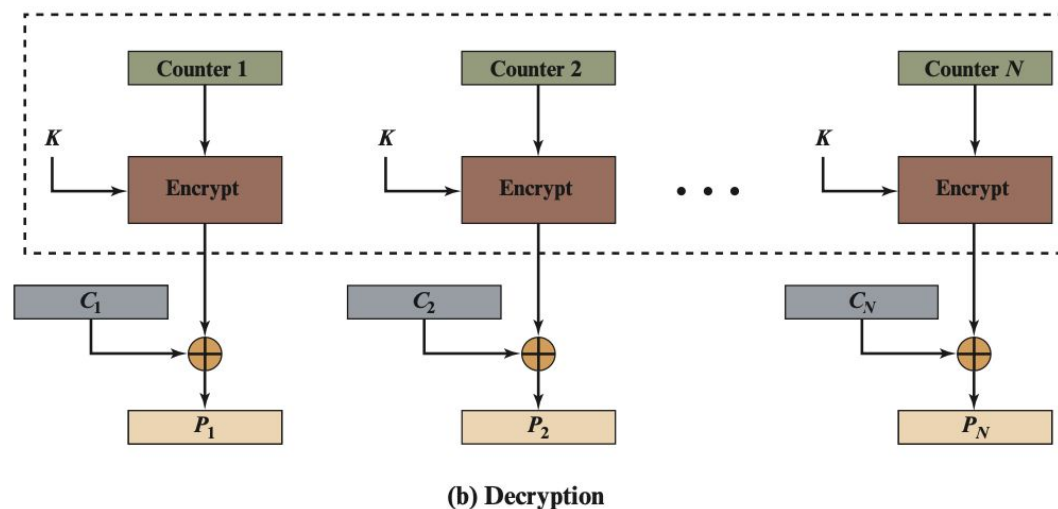
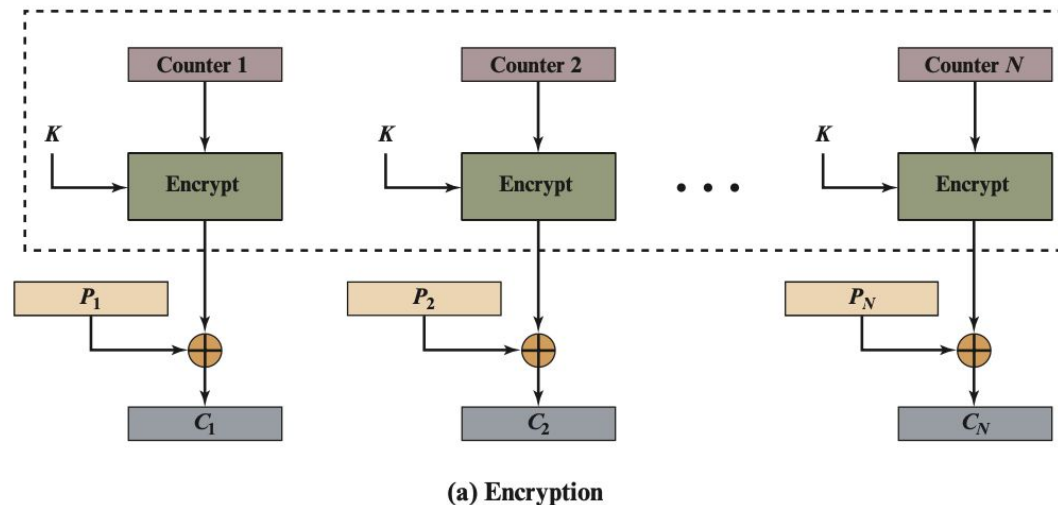
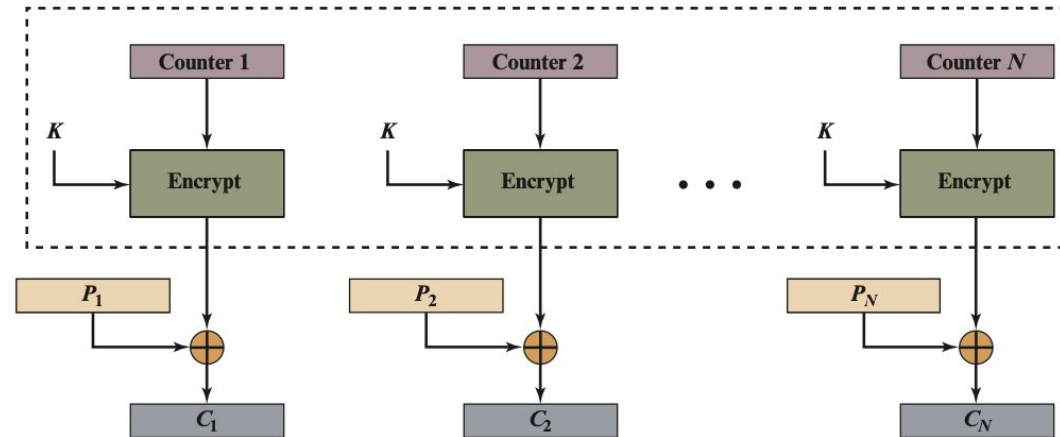


Figure 7.7 Counter (CTR) Mode

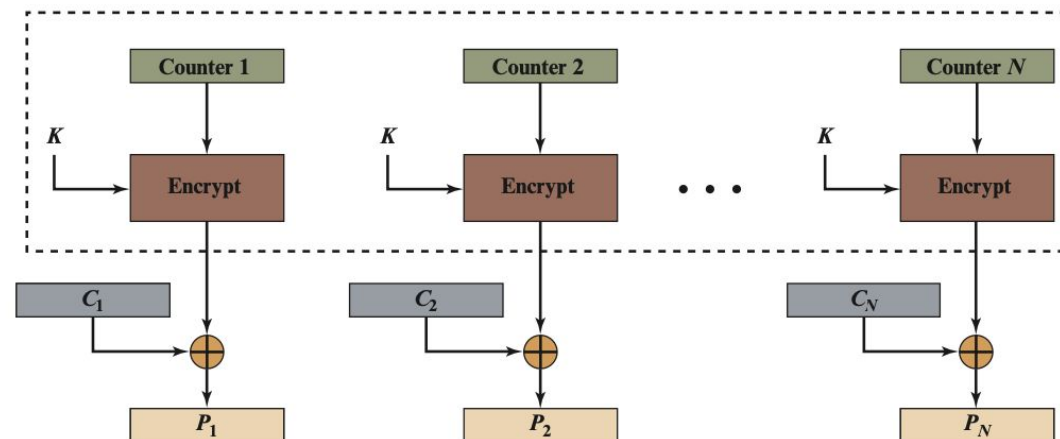
# COUNTER MODE (CTR)

**Hardware efficiency:** Unlike the three chaining modes, encryption in CTR mode can be done in **parallel** on multiple blocks of plaintext.

**Preprocessing:** The execution of the underlying encryption algorithm **does not depend on** input of the plaintext or ciphertext.



(a) Encryption



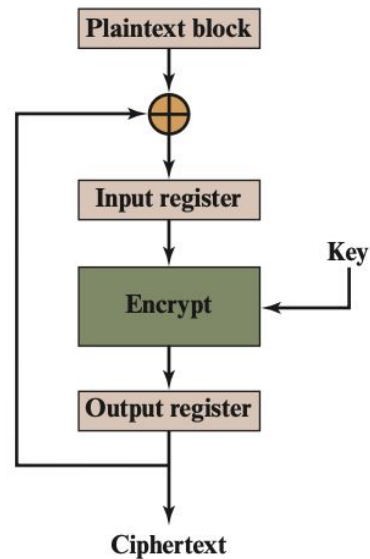
(b) Decryption

Figure 7.7 Counter (CTR) Mode

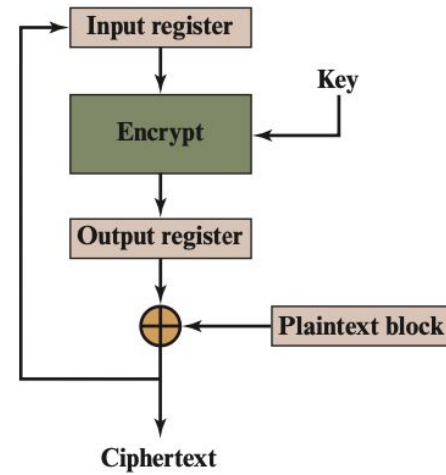


**Table 7.1** Block Cipher Modes of Operation

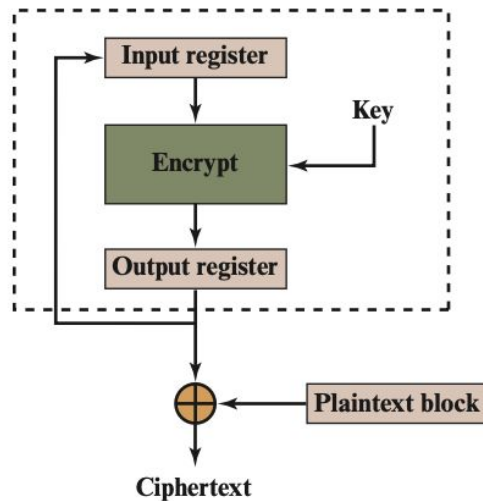
| Mode                        | Description                                                                                                                                                                                                | Typical Application                                                                                                                        |
|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|
| Electronic Codebook (ECB)   | Each block of plaintext bits is encoded independently using the same key.                                                                                                                                  | <ul style="list-style-type: none"><li>• Secure transmission of single values (e.g., an encryption key)</li></ul>                           |
| Cipher Block Chaining (CBC) | The input to the encryption algorithm is the XOR of the next block of plaintext and the preceding block of ciphertext.                                                                                     | <ul style="list-style-type: none"><li>• General-purpose block-oriented transmission</li><li>• Authentication</li></ul>                     |
| Cipher Feedback (CFB)       | Input is processed $s$ bits at a time. Preceding ciphertext is used as input to the encryption algorithm to produce pseudorandom output, which is XORed with plaintext to produce next unit of ciphertext. | <ul style="list-style-type: none"><li>• General-purpose stream-oriented transmission</li><li>• Authentication</li></ul>                    |
| Output Feedback (OFB)       | Similar to CFB, except that the input to the encryption algorithm is the preceding encryption output, and full blocks are used.                                                                            | <ul style="list-style-type: none"><li>• Stream-oriented transmission over noisy channel (e.g., satellite communication)</li></ul>          |
| Counter (CTR)               | Each block of plaintext is XORed with an encrypted counter. The counter is incremented for each subsequent block.                                                                                          | <ul style="list-style-type: none"><li>• General-purpose block-oriented transmission</li><li>• Useful for high-speed requirements</li></ul> |



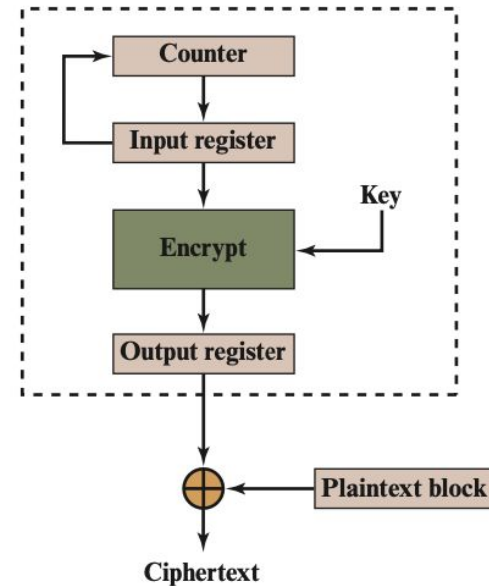
(a) Cipher block chaining (CBC) mode



(b) Cipher feedback (CFB) mode



(c) Output feedback (OFB) mode



(d) Counter (CTR) mode

**Figure 7.8** Feedback Characteristic of Modes of Operation